

CS 171: Discussion Section 4 (9/29)

1 Pseudorandom Functions

Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function. Prove whether the candidate following candidate is pseudorandom or not:

$$f'_{(k_1, k_2)}(x) = f_{k_1}(x) \parallel f_{k_2}(x).$$

Solution Yes. We will show this via a hybrid argument using the following hybrids:

- $\text{Hyb}_0(x) := f_{k_1}(x) \parallel f_{k_2}(x).$
- $\text{Hyb}_1(x) := R_1(x) \parallel f_{k_2}(x)$ where R_1 is a random function.
- $\text{Hyb}_2(x) := R_1(x) \parallel R_2(x)$ where R_1, R_2 are random functions.

Following the definition of a pseudorandom function given by Katz-Lindell (Definition 3.25 in the 2nd edition), it suffices to show that for any PPT distinguisher D

$$|\Pr[D^{\text{Hyb}_0(\cdot)}(1^n) = 1] - \Pr[D^{\text{Hyb}_2(\cdot)}(1^n) = 1]| \leq \text{negl}(n)$$

since $\text{Hyb}_1(\cdot)$ has the same distribution as $f'_{(k_1, k_2)}$ when $k_1, k_2 \xleftarrow{\$} \{0, 1\}^n$ and $\text{Hyb}_2(\cdot)$ has the same distribution as selecting a function uniformly at random from Func_{2n} . We use the intermediate hybrid $\text{Hyb}_1(x)$ to establish the result from two claims.

Claim 1.1. *For any PPT distinguisher D ,*

$$|\Pr[D^{\text{Hyb}_0(\cdot)}(1^n) = 1] - \Pr[D^{\text{Hyb}_1(\cdot)}(1^n) = 1]| \leq \text{negl}(n)$$

Proof. Assume for the sake of contradiction that there exists an adversary D that distinguishes between oracle access to the functions in Hyb_1 and Hyb_0 with non-negligible advantage. We will construct an adversary D' against the pseudorandomness property of f_{k_1} . D' samples k_2 randomly. D' runs D and on any oracle query x made by D , D' queries its own oracle on x to get y . D' returns $y \parallel f_{k_2}(x)$. It is easy to see that D' runs in polynomial time since D runs in poly time.

If D' has been given access to the f_{k_1} oracle then its oracle responses are identically distributed to $\text{Hyb}_0(\cdot)$. Otherwise the responses are distributed identically to $\text{Hyb}_1(\cdot)$. Hence the advantage of D' distinguishing oracle access to $f_{k_1}(\cdot)$ and the random oracle is same as the advantage of D' in distinguishing between oracle access to Hyb_0 from Hyb_1 , and this is assumed to be non-negligible. A contradiction. $\square$

Claim 1.2. *For any PPT distinguisher D ,*

$$|\Pr[D^{\text{Hyb}_1(\cdot)}(1^n) = 1] - \Pr[D^{\text{Hyb}_2(\cdot)}(1^n) = 1]| \leq \text{negl}(n)$$

Proof. Assume for the sake of contradiction that there exists an adversary D that distinguishes between oracle access to the functions in Hyb_1 and Hyb_2 with non-negligible advantage. We will construct an adversary D' against the pseudorandomness property of f_{k_2} . D'

runs D and on any oracle query x made by D , D' queries its own oracle on x to get y . D' returns $r \parallel y$ where r is a uniformly chosen random string. It stores (x, r) and the next time D queries x , it uses the stored value to answer this query. It is easy to see that D' runs in polynomial time since D runs in poly time.

If D' has been given access to the f_{k_2} oracle then its oracle responses are identically distributed to $\text{Hyb}_1(\cdot)$. Otherwise the responses are distributed identically to $\text{Hyb}_2(\cdot)$. Hence the advantage of D' distinguishing oracle access to $f_{k_2}()$ and the random oracle is same as the advantage of D' in distinguishing between oracle access to Hyb_1 from Hyb_2 , and this is assumed to be non-negligible. A contradiction. $\square$

Putting these two claims together, for any PPT distinguisher D ,

$$\begin{aligned} |\Pr[D^{\text{Hyb}_1(\cdot)}(1^n) = 1] - \Pr[D^{\text{Hyb}_2(\cdot)}(1^n) = 1]| &\leq |\Pr[D^{\text{Hyb}_0(\cdot)}(1^n) = 1] - \Pr[D^{\text{Hyb}_1(\cdot)}(1^n) = 1]| \\ &\quad + |\Pr[D^{\text{Hyb}_1(\cdot)}(1^n) = 1] - \Pr[D^{\text{Hyb}_2(\cdot)}(1^n) = 1]| \\ &\leq \text{negl}(n) + \text{negl}(n) = \text{negl}(n), \end{aligned}$$

establishing the desired result. $\square$

2 CPA-Secure Encryption

Let $(\text{Gen}, \text{Enc}, \text{Dec})$ be a CPA-secure encryption scheme. Below, we will construct another encryption scheme and prove that it is also CPA-secure.

In the encryption scheme below, let the message m belong to $\{0, 1\}^n$.

- $\text{Gen}_1(1^n)$: Sample the key as follows: $k \leftarrow \text{Gen}(1^n)$.
- $\text{Enc}_1(k, m)$: Sample $r \leftarrow \{0, 1\}^n$ uniformly at random. Then compute $c_0 := \text{Enc}(k, r)$ and $c_1 := r \oplus m$. Output the ciphertext $c = (c_0, c_1)$.
- $\text{Dec}_1(k, (c_0, c_1))$: **Unspecified**

1. Fill in the decryption algorithm so that every ciphertext is decrypted correctly.

$\text{Dec}_1(k, (c_0, c_1))$:

2. Prove that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ satisfies CPA security.

Solution

1. $\text{Dec}_1(k, (c_0, c_1))$: Compute $r' := \text{Dec}(k, c_0)$ and then compute $m' := r' \oplus c_1$. Output m' .

2. **Note:** In the solution below, we've included a lot more detail than students would be expected to give on e.g. an exam. For ease of reading, we've marked in gray the sections that can be skipped if you are just skimming the solution.

1. Let's define two hybrids that are identical, except in 2.1 and 2.2 below.

- H_0 is the CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$. Also, let $\mathcal{A}$ be the adversary in this game.

Here's the hybrid in more detail. (This level of detail is optional, but it can be very helpful to the reader of your proofs).

- (a) **Setup:** The challenger samples $k \leftarrow \text{Gen}_1(1^n)$.
- (b) **Phase I queries:** $\mathcal{A}$ sends the challenger a message, and the challenger responds with

$$c = (\text{Enc}(k, r), (r \oplus m))$$

where $r \leftarrow \{0, 1\}^n$. $\mathcal{A}$ can repeat this step many times.

- (c) **Challenge:** $\mathcal{A}$ outputs two messages m_0, m_1 . The challenger samples a bit $b \leftarrow \{0, 1\}$, and sends $\mathcal{A}$ the encryption c^* of m_b :

$$c^* = \text{Enc}(k, r), (r \oplus m_b) \tag{2.1}$$

where r is sampled uniformly at random.

- (d) **Phase II queries:** Work the same as phase I queries.
- (e) **Output:** $\mathcal{A}$ outputs a bit b' . The output of the hybrid is 1 if $b = b'$ and 0 otherwise.
- H_1 is the same as H_0 except the challenge ciphertext c^* is $(\text{Enc}(k, 0^n), (r \oplus m_b))$.

Here's the hybrid in more detail, with any change from Hybrid 0 underlined:

- (a) **Setup:** The challenger samples $k \leftarrow \text{Gen}_1(1^n)$.
- (b) **Phase I queries:** $\mathcal{A}$ sends the challenger a message, and the challenger responds with

$$c = (\text{Enc}(k, r), (r \oplus m))$$

where $r \leftarrow \{0, 1\}^n$. $\mathcal{A}$ can repeat this step many times.

- (c) **Challenge:** $\mathcal{A}$ outputs two messages m_0, m_1 . The challenger samples a bit $b \leftarrow \{0, 1\}$, and sends $\mathcal{A}$ the encryption c^* of m_b :

$$c^* = \text{Enc}(k, \underline{0^n}), (r \oplus m_b) \tag{2.2}$$

where r is sampled uniformly at random.

- (d) **Phase II queries:** Work the same as phase I queries.
- (e) **Output:** $\mathcal{A}$ outputs a bit b' . The output of the hybrid is 1 if $b = b'$ and 0 otherwise.

2.

Claim 2.1. *If $(\text{Gen}, \text{Enc}, \text{Dec})$ is CPA secure, then for any adversary $\mathcal{A}$, $|\Pr[H_0 \rightarrow 1] - \Pr[H_1 \rightarrow 1]|$ is negligible.*

Proof. (a) Assume toward contradiction that for some adversary $\mathcal{A}$, $|\Pr[H_0 \rightarrow 1] - \Pr[H_1 \rightarrow 1]|$ is non-negligible. Then we will construct an adversary $\mathcal{B}$ that breaks the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$.

- (b) **Notation:** We'll use (M_0, M_1, B, C^*) to denote some of the variables in the CPA security game in which $\mathcal{B}$ is playing so that they don't get mixed up with (m_0, m_1, b, c^*) from the hybrids above.
- (c) $\mathcal{B}$ is designed to simulate one of the hybrids, H_B , where $B \in \{0, 1\}$ is chosen by the CPA challenger.

At the beginning of the CPA security game, the challenger samples $k \leftarrow \text{Gen}(1^n)$, which serves to simulate step a of the hybrids.

Description of $\mathcal{B}$:

- i. In phase I of the CPA game, $\mathcal{B}$ will simulate step b of the hybrids. This entails running $\mathcal{A}$, and when $\mathcal{A}$ outputs an encryption query m , $\mathcal{B}$ will sample $r \leftarrow \{0, 1\}^n$ and send $\mathcal{A}$ the response: $c = (\text{Enc}(k, r), (r \oplus m))$. This requires $\mathcal{B}$ to make a query to $\text{Enc}(k, \cdot)$.
- ii. In the challenge phase:

- A. $\mathcal{B}$ samples an $r \leftarrow \{0, 1\}^n$ and outputs two challenge messages, $M_0 = r, M_1 = 0^n$, and receives in response either $C^* = \text{Enc}(k, r)$ (when $B = 0$) or $C^* = \text{Enc}(k, 0^n)$ (when $B = 1$).
- B. Next, C^* allows $\mathcal{B}$ to simulate step c of the hybrids. When $\mathcal{A}$ outputs its messages (m_0, m_1) in step c, $\mathcal{B}$ will sample a bit $b \leftarrow \{0, 1\}$ and respond to $\mathcal{A}$ with

$$C^*, (r \oplus m_b)$$

Note that if $C^* = \text{Enc}(k, r)$, then $\mathcal{B}$ has simulated step c of H_0 , but if $C^* = \text{Enc}(k, 0^n)$, then $\mathcal{B}$ has simulated step c of H_1 .

- iii. In phase II, $\mathcal{B}$ will simulate steps d and e of the hybrids. The output of the hybrid is a bit, which $\mathcal{B}$ outputs as well.

- (d) As we argued above, $\mathcal{B}$ simulates H_B . Therefore,

$$\Pr[\mathcal{B} \rightarrow 1 | B = 0] = \Pr[H_0 \rightarrow 1]$$

$$\Pr[\mathcal{B} \rightarrow 1 | B = 1] = \Pr[H_1 \rightarrow 1]$$

$$\begin{aligned} \left| \Pr[\mathcal{B} \rightarrow 1 | B = 0] - \Pr[\mathcal{B} \rightarrow 1 | B = 1] \right| &= \left| \Pr[H_0 \rightarrow 1] - \Pr[H_1 \rightarrow 1] \right| \\ &= \text{non-negligible}(n) \end{aligned}$$

This means that $\mathcal{B}$ breaks the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$.

- (e) However, the problem states that $(\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, so we've arrived at a contradiction. This means that our initial assumption is false, and in reality:

$$\left| \Pr[H_0 \rightarrow 1] - \Pr[H_1 \rightarrow 1] \right| = \text{negl}(n)$$

□

3.

Claim 2.2. $\Pr[H_1 \rightarrow 1] = \frac{1}{2}$

Proof. Recall that in H_1 , the challenge ciphertext is $c^* = (\text{Enc}(k, 0^n), (r \oplus m_b))$. This ciphertext gives the adversary no information about b because m_b is masked by a uniformly random r .

The only place where r and b appear in the ciphertext is in $r \oplus m_b$. Furthermore, $r \oplus m_b$ is a random string that is independent of b because r is uniformly random. Therefore, the adversary's probability of guessing b correctly (i.e. the probability that $b' = b$) is exactly $\frac{1}{2}$. □

- 4. Putting everything together, we have that:

$$\begin{aligned} \Pr[H_0 \rightarrow 1] &\leq \Pr[H_1 \rightarrow 1] + \left| \Pr[H_0 \rightarrow 1] - \Pr[H_1 \rightarrow 1] \right| \\ &= \frac{1}{2} + \text{negl}(n) \end{aligned}$$

Since H_0 is the CPA security experiment for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$, this means that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ is CPA-secure.

□

A Flawed Approach

Here is a similar approach that doesn't quite work. It's useful to understand why not.

Claim 2.3. *If $(\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, then $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ is also CPA-secure.*

Proof. 1. Assume toward contradiction that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ does not satisfy CPA security. Then there is an adversary $\mathcal{A}$ that wins the CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ with probability $\frac{1}{2} + \text{non-negligible}(n)$. We will use $\mathcal{A}$ to construct an adversary $\mathcal{B}$ that breaks the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$, which is a contradiction. Then we can conclude that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ does satisfy CPA security.

2. Description of $\mathcal{B}$:

- (a) $\mathcal{B}$ runs $\mathcal{A}$ and responds to any queries made by $\mathcal{A}$. When $\mathcal{A}$ outputs a phase-I message m , $\mathcal{B}$ samples $r \leftarrow \{0, 1\}^n$ and computes $\text{Enc}(k, r)$ by making a phase-I query to its own oracle $\text{Enc}(k, \cdot)$. Then $\mathcal{B}$ computes $c := \text{Enc}(k, r), (r \oplus m)$ and sends c to $\mathcal{A}$.
 - (b) In the challenge phase of $\mathcal{A}$'s CPA game, $\mathcal{A}$ outputs two messages (m_0, m_1) . $\mathcal{B}$ samples a new $r \leftarrow \{0, 1\}^n$ and outputs $(r, 0^n)$ as its challenge messages. $\mathcal{B}$ receives a ciphertext C^* from the challenger, which is either $\text{Enc}(k, r)$ or $\text{Enc}(k, 0^n)$. $\mathcal{B}$ sends $(C^*, (r \oplus m_0))$ to $\mathcal{A}$.
 - (c) In phase II of $\mathcal{B}$'s CPA game, $\mathcal{B}$ will simulate phase II of $\mathcal{A}$'s CPA game. This works the same way as phase I.
 - (d) Finally, $\mathcal{A}$ will output a bit b' , which $\mathcal{B}$ also outputs.
3. When $C^* = \text{Enc}(k, r)$, then $\mathcal{A}$ receives $(\text{Enc}(k, r), (r \oplus m_0))$, which is a valid encryption of m_0 under Enc_1 . **In this case, the probability that $\mathcal{B}$ wins the CPA game is $\frac{1}{2} + \delta(n)$, where δ is some non-negligible function (*This analysis is incorrect*).**
 4. When $C^* = \text{Enc}(k, 0^n)$, then $\mathcal{A}$ receives $\text{Enc}(k, 0^n), (r \oplus m_0)$, which is not a valid encryption of m_0 . In fact, it gives no information about m_0 because m_0 is masked by a uniformly random string r . Therefore, in this case, the probability that $\mathcal{B}$ wins the CPA game is exactly $\frac{1}{2}$.
 5. In summary, the probability that $\mathcal{B}$ wins the CPA game is:

$$\frac{1}{2} \cdot \left(\frac{1}{2} + \delta(n) \right) + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} + \frac{\delta}{2}$$

which is still non-negligibly greater than $\frac{1}{2}$. □

To illustrate the problem with this analysis, let's construct an adversary $\mathcal{A}$ that is purposefully unhelpful to $\mathcal{B}$.

Description of $\mathcal{A}$: Let's say that $\mathcal{A}$ can decrypt any ciphertext generated by $\text{Enc}_1(k, \cdot)$. When $\mathcal{A}$ receives its challenge ciphertext, let $\mathcal{A}$ decrypt the ciphertext to get m^* . If $m^* \neq m_1$, then $\mathcal{A}$ samples $b' \leftarrow \{0, 1\}$ uniformly at random and outputs b' . If $m^* = m_1$, then $\mathcal{A}$ outputs $b' = 1$.

Next, the claims below show that $\mathcal{A}$ breaks the CPA security of $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$, but $\mathcal{B}$ does not break the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$. Recall that the goal of the proof was to show that if $\mathcal{A}$ breaks the CPA security of $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$, then $\mathcal{B}$ breaks the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$, so the proof is unsuccessful.

Claim 2.4. $\mathcal{A}$ wins the CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ with probability $\frac{3}{4}$.

Proof. If the CPA challenger encrypted m_0 , then $\mathcal{A}$ outputs the correct answer ($b' = 0$) with probability $\frac{1}{2}$. If the CPA challenger encrypted m_1 , then $\mathcal{A}$ outputs the correct answer ($b' = 1$) with probability 1. Therefore, $\mathcal{A}$ wins the CPA security game with probability

$$\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 = \frac{3}{4}$$

□

Claim 2.5. $\mathcal{B}$ wins the CPA security game for $(\text{Gen}, \text{Enc}, \text{Dec})$ with probability $\frac{1}{2}$.

Proof. If the CPA challenger encrypted M_0 , then $\mathcal{B}$ outputs the correct answer ($B' = 0$) with probability $\frac{1}{2}$. If the CPA challenger encrypted M_1 , then $\mathcal{B}$ outputs the correct answer ($B' = 1$) with probability $\frac{1}{2}$. Therefore, $\mathcal{B}$ wins the CPA security game with probability $\frac{1}{2}$. □

Another (valid) solution

This solution is a little simpler than the first solution given above. Rather than using $\mathcal{A}$ to distinguish between a valid and an invalid ciphertext, we will use $\mathcal{A}$ to distinguish between two valid ciphertexts. This solution is due to an anonymous student – thank-you to them!

Claim 2.6. If $(\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, then $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ is also CPA-secure.

Proof. 1. Assume toward contradiction that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ does not satisfy CPA security. Then there is an adversary $\mathcal{A}$ that wins the CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ with probability $\frac{1}{2}$ plus non-negligible. We will use $\mathcal{A}$ to construct an adversary $\mathcal{B}$ that breaks the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$, which is a contradiction. Then we can conclude that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ does satisfy CPA security.

2. Description of $\mathcal{B}$:

- (a) $\mathcal{B}$ runs $\mathcal{A}$ and responds to any queries made by $\mathcal{A}$. When $\mathcal{A}$ outputs a phase-I message m , $\mathcal{B}$ samples $r \leftarrow \{0, 1\}^n$ and computes $\text{Enc}(k, r)$ by making a phase-I query to its own oracle $\text{Enc}(k, \cdot)$. Then $\mathcal{B}$ computes $c := (\text{Enc}(k, r), (r \oplus m))$ and sends c to $\mathcal{A}$.
- (b) In the challenge phase of $\mathcal{A}$'s CPA game, $\mathcal{A}$ outputs two messages (m_0, m_1) . $\mathcal{B}$ samples a ciphertext $c_1^* \leftarrow \{0, 1\}^n$. Then it computes $r_0 := c_1^* \oplus m_0$ and $r_1 := c_1^* \oplus m_1$. Then $\mathcal{B}$ outputs (r_0, r_1) as its challenge messages.
The CPA challenger samples $B \leftarrow \{0, 1\}$ and sends $\text{Enc}(k, r_B)$ to $\mathcal{B}$. Then $\mathcal{B}$ sends

$$(\text{Enc}(k, r_B), c_1^*)$$

to $\mathcal{A}$.

- (c) In phase II of $\mathcal{B}$'s CPA game, $\mathcal{B}$ will simulate phase II of $\mathcal{A}$'s CPA game. This works the same way as phase I.
 - (d) Finally, $\mathcal{A}$ will output a bit b' , which $\mathcal{B}$ also outputs.
3. Note that for either value of B , $(\text{Enc}(k, r_B), c_1^*)$ is a valid encryption of m_B under $\text{Enc}_1(k, \cdot)$. This is because $c_1^* = (r_B \oplus m_B)$. Furthermore, r_B is uniformly random and independent of (m_0, m_1, B) . So $\mathcal{B}$ has correctly simulated the CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$.
 4. If $\mathcal{A}$ wins the simulated CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$, by correctly guessing B , then $\mathcal{B}$ wins the CPA security game for $(\text{Gen}, \text{Enc}, \text{Dec})$. Therefore, $\mathcal{A}$'s success probability in the CPA security game for $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ equals $\mathcal{B}$'s success probability in the CPA security game for $(\text{Gen}, \text{Enc}, \text{Dec})$. If $\mathcal{A}$ breaks the CPA security of $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ (by winning the corresponding CPA game with probability $\frac{1}{2} + \text{non-negligible}(n)$), then $\mathcal{B}$ breaks the CPA security of $(\text{Gen}, \text{Enc}, \text{Dec})$.
 5. We know that $(\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, so our initial assumption must be false, and in truth, $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ is also CPA-secure.

□